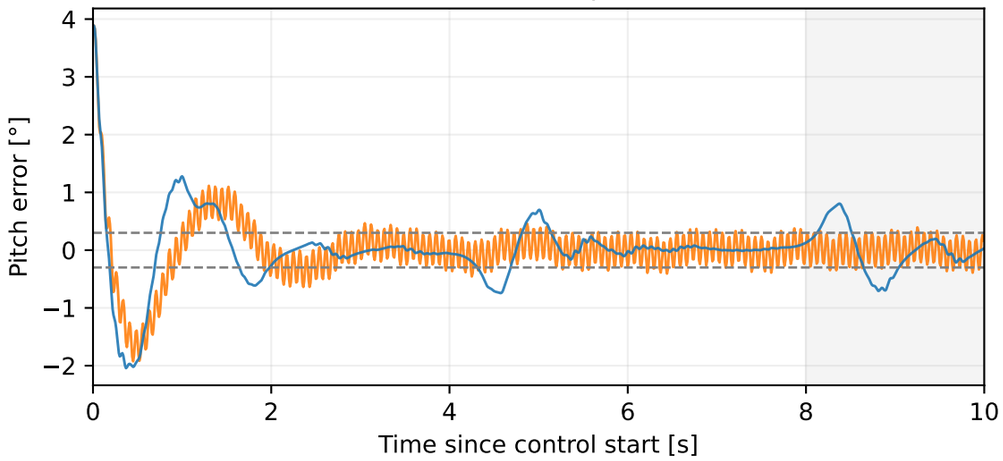
MuJoCo — baseline LQR versus GA-tuned LQR (Q and R)

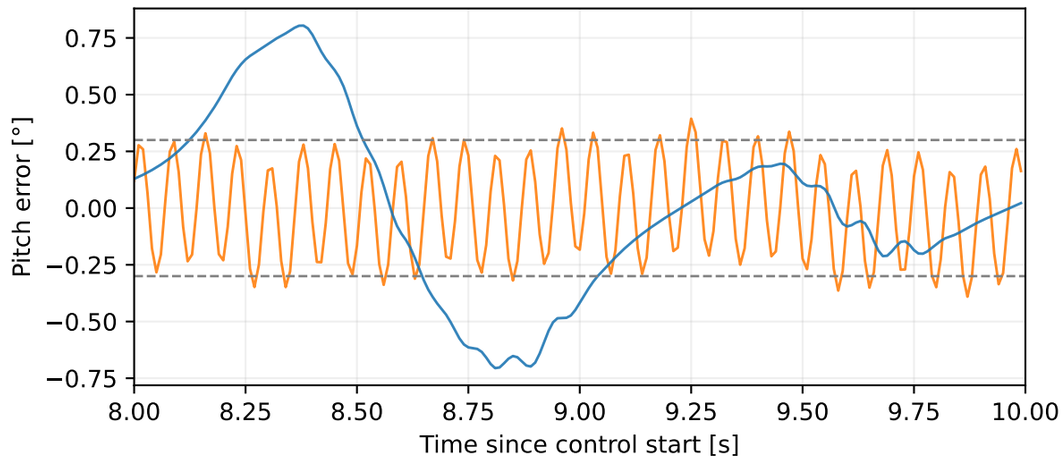
validation_00 | single-trial cost: 0.6494 → 0.7973 | all-scenario cost: 1.0811 → 1.0806

Baseline LQR GA-tuned LQR

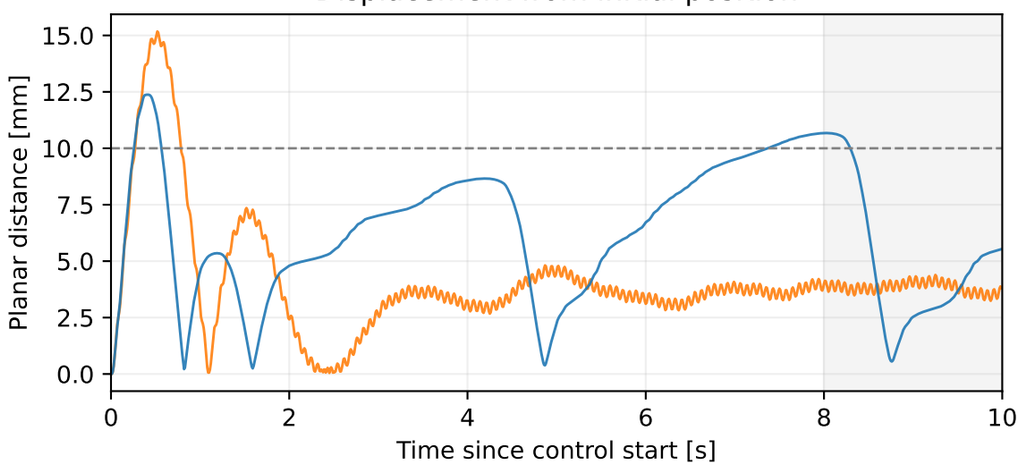
Tilt relative to equilibrium



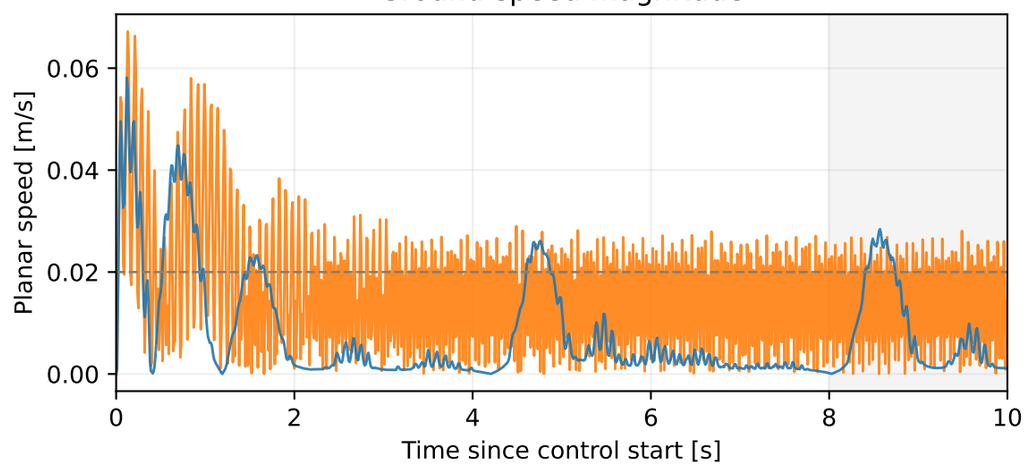
Final 2 s detail



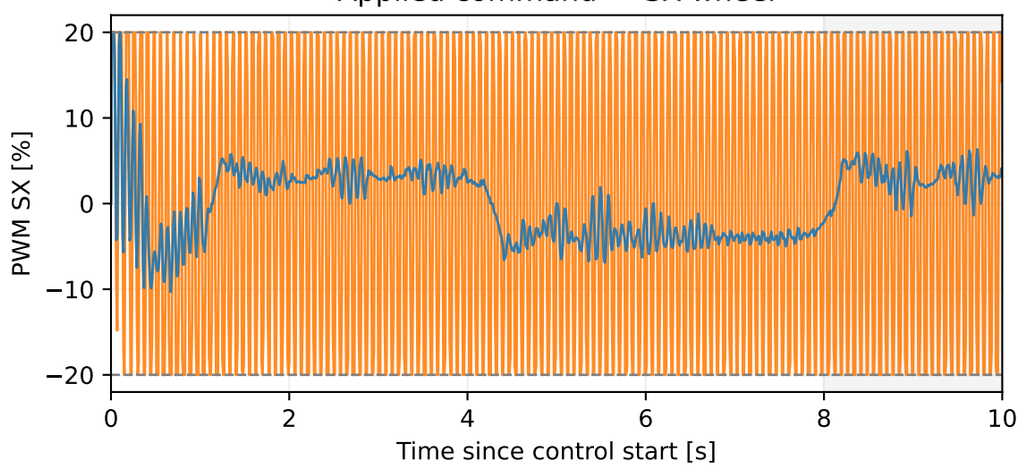
Displacement from initial position



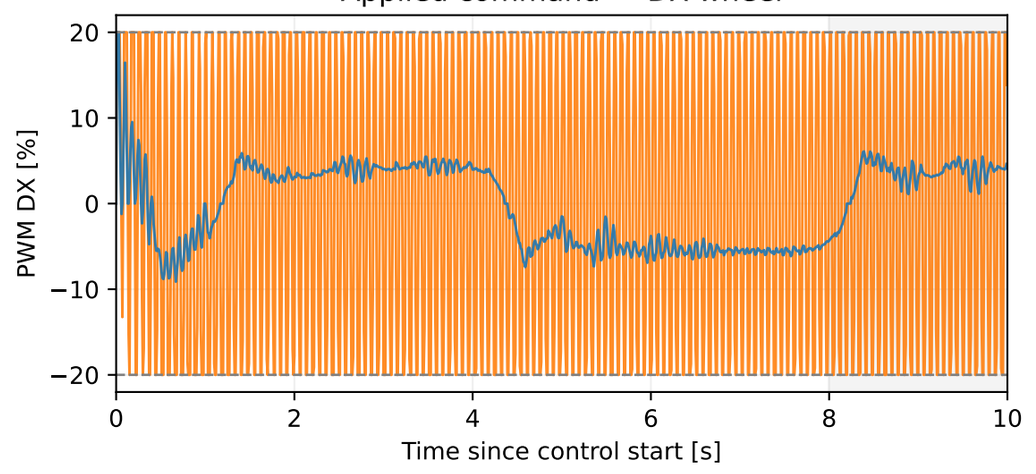
Ground-speed magnitude



Applied command — SX wheel



Applied command — DX wheel



Same SX/DX and Stribeck model, sensor seed and protocol. Gray: final assessment. Dashed: thresholds or PWM limits.
baseline: duration completed, settling criteria not met; ga: duration completed, settling criteria not met